for Noun Phrase, VP stands for Verb Phrase, and DT stands for Determiner. This can be considered as a labelling problem in machine learning where each word is labelled with its associated part of speech tag. Instead of considering each word in isolation, using the context around the word will help disambiguate the correct tag for each word. For example, consider Sentence 1: "Please give me the correct change." Here the word 'change' is used as a noun phrase and not as a verb phrase. In Sentence 2: "You will never change your behaviour," the word 'change' is used as a verb phrase. The context of the surrounding tags in Sentence 1 helps to disambiguate between the two possible tags <Noun Phrase> and <Verb Phrase>. Hence, parts of speech tagging is typically framed as a sequence labelling problem. Two popular techniques available for sequence labelling are Hidden Markov Models (HMM) and Conditional Random Fields (CRF). Can you distinguish between HMM and CRF? For the above problem of PoS tagging, would you use HMM or CRF? Can you provide reasons for your choice?

(9) Named Entity Recognition (NER) is a major challenge in natural language processing, where named entities in the text, namely, the person, location, organisation, etc, need to be recognised automatically. While machine learning techniques are typically used for NER, many of the cases can be captured by simple rule based techniques. You are asked to write a C program to recognise persons' names in a text document. Persons' names would have the first letter of the word as a capitalised letter of the English alphabet, followed by lower case letters. It is also possible that persons' names can be multi-word, such as John Smith. Your program should recognise both single and multi-word persons' names.

(10) Automatic summarisation techniques are used to extract summaries of large text documents. Can you explain the difference between extractive summarisation and abstractive summarisation techniques?

(11) You are given a large text file. The file may contain duplicate lines. You are asked to write a program to remove duplicate lines from the text file. (a) Assume that you can create a new text file which contains the text with the duplicate lines removed. (b) You are asked to remove the lines in place in the original file itself without creating another file.

(12) You are given a large text document along with the PoS list for the whole document, where the PoS list for each word specifies the positions in the document where that particular word appears. The PoS list structure can be considered as a list of arrays, with one array for each word in the document. For example, consider the document as containing the following two sentences: "Ram ate the apple. The apple was sweet."
The PoS list would look as follows:
Ram -> [1]. ate -> [2]. The -> [3,5]. Apple-> [4,6]. Was -> [7]. Sweet -> [8].
The index positions in the document are numbered from 1. You are given a list of words and asked to find the smallest length sequence in the document in which all the words in your list are present. For example, if you are given the list <apple, sweet>, the answer would be [6-8]. The answer sequence should be specified in terms of the start and end positions of the sequence in the document. Can you write a C program to solve the above problem?

(13) We are all very familiar with Facebook's 'Like' feature. What kind of data structure would you use to implement such a feature? What would be the space complexity? For a given user, if you wanted to provide recommendations of posts which have been liked by friends, how would you support this efficiently?

(14) You are given two strings, S1 and S2, of unequal lengths N1 and N2. You are asked to transform the string S1 into S2 by doing a minimum number of (a) insertions of a character and (b) deletions of a character. Can you write a C program to solve this problem?

(15) Let us consider a binary classification problem for which you are asked to label emails as spam or non-spam. You are given a large enough training data set (100,000 emails), where the emails are already classified as spam or non-spam. Using this training data, you are asked to train a binary classifier which can mark an incoming email as spam or non-spam. However, there is one problem with the training data that you have been given. Among the 100,000 training data emails, only 100 of them are labelled as spam. All other training data are instances of the non-spam category. What would be the problem if you train your model with such an unbalanced training data set? How would you overcome this issue?

If you have any favourite programming questions/ software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Till we meet again next month, wishing all our readers, a wonderful new year ahead! END

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